

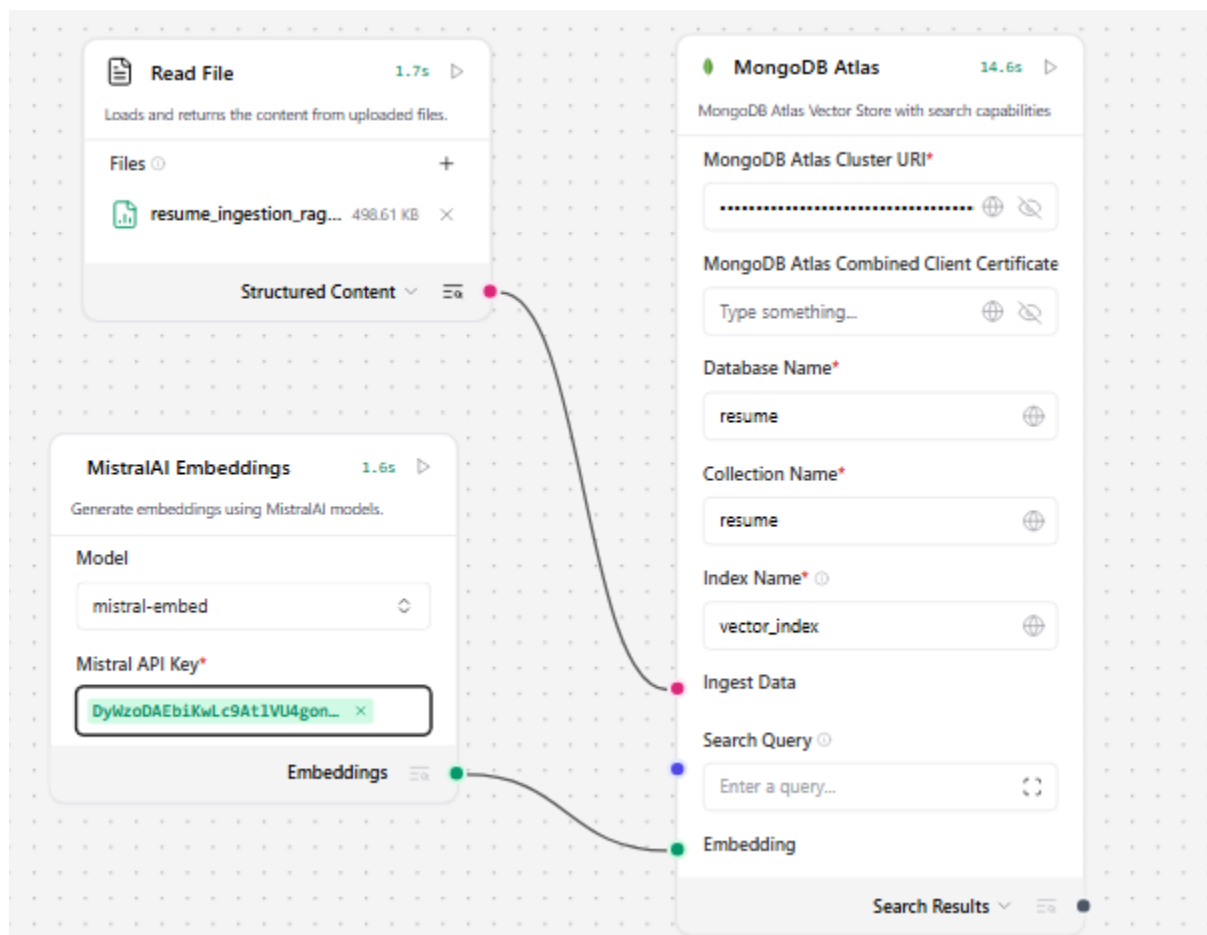
CREATE NEW DB COLLECTION AND CREATE BM AND VECTOR SEARCH INDEX

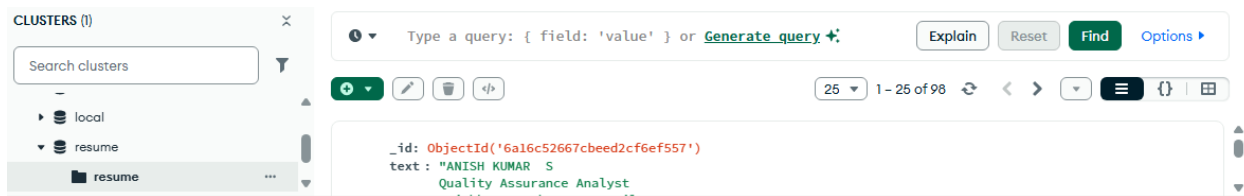
Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Docu
resume	resume	BM25_Index	READY View status details	✓	search	"certifications", "currentOrganisation", "currentTitle", "educationLevel", "email", + 15 more	0 (09
resume	resume	vector_index	READY View status details	✓	vectorSearch	"embedding", "id", "name", "email", "phone" + 14 more	0 (09

[Learn more about Atlas Search](#)

[Learn more about Atlas Vector Search](#)

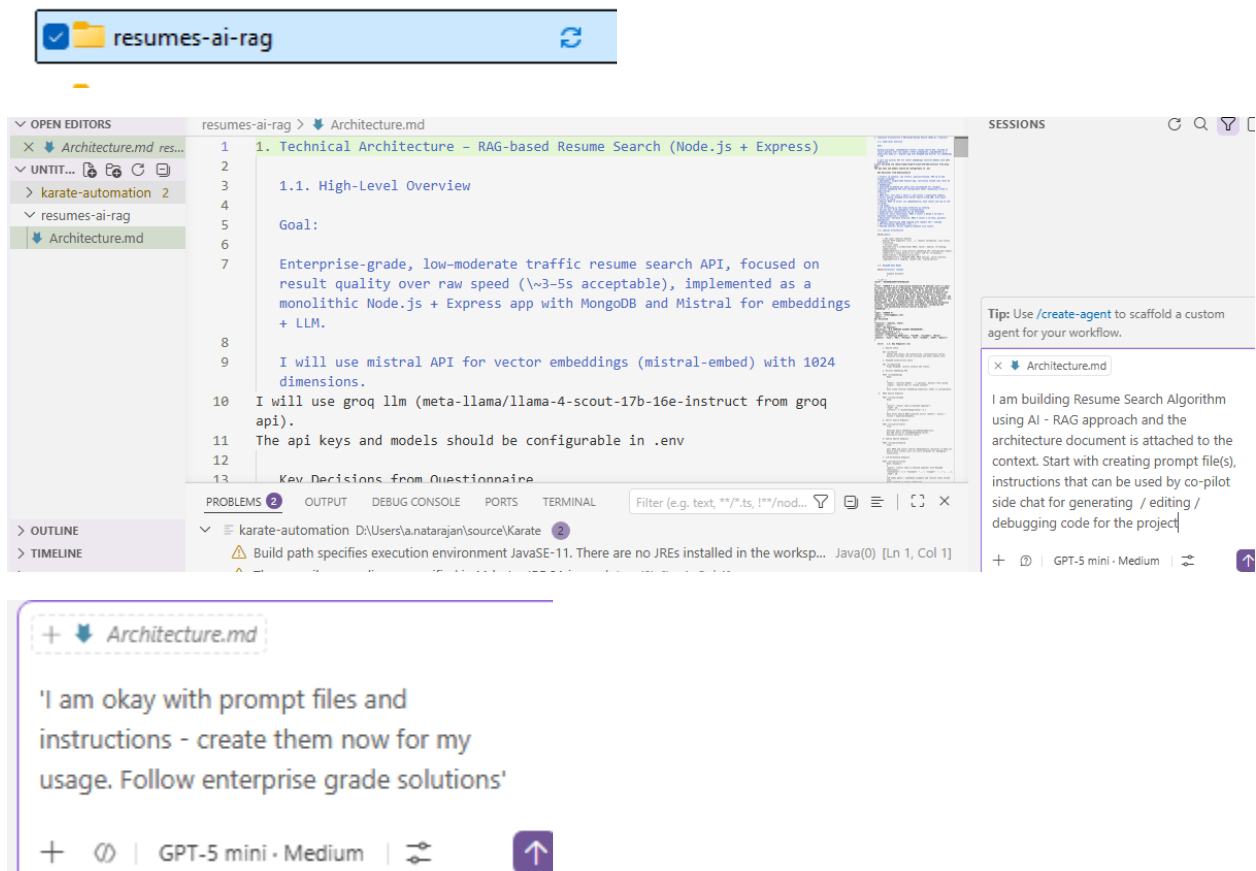
USING LANGFLOW SHARED FLOW INGESTED 98 RESUMES



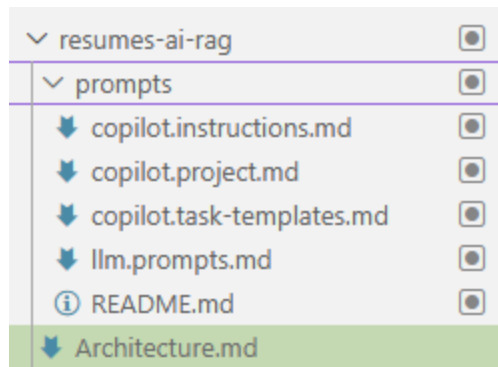


LET'S COMMENCE STEPS FOR RAG RETRIEVAL PIPELINE

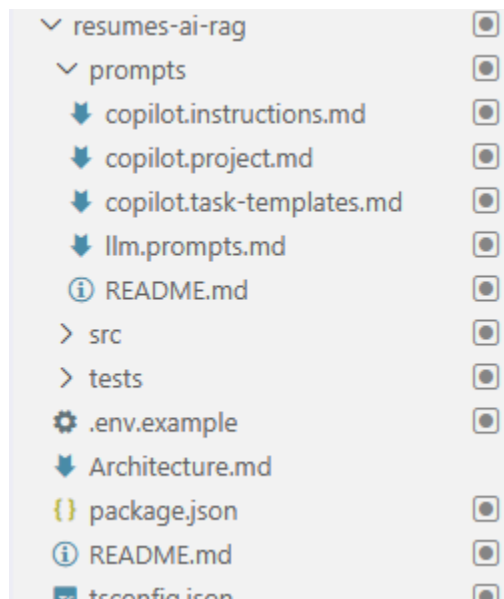
Create a new folder 'resumes-ai-rag'



Prompt templates are created



LETS CREATE FOLDER STRUCTURE



LETS RUN NPM INSTALL

```
up to date, audited 282 packages in 2s

56 packages are looking for funding
  run `npm fund` for details

5 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
npm audit fix --force
```

Lets Update your mongodb connection(db_url, db_name, collection_name, bm25_index, vector_index) in the .env and save it .

```

MONGO_DB_NAME=resumes_db
MISTRAL_API_KEY=
MISTRAL_EMBED_MODEL=mistral-embed
MISTRAL_LLM_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
LOG_LEVEL=info
REQUEST_SIZE_LIMIT=1048576
BM25_INDEX=BM25_index
VECTOR_INDEX=vector_index

```

START THE SERVER AND PERFORM HEALTHCHECK

HTTP <http://localhost:3000/v1/health> Save Share

GET <http://localhost:3000/v1/health> Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results 200 OK 10 ms 399 B ...

{ } JSON Preview Visualize

```

1 {
2   "name": "resumes-ai-rag",
3   "version": "0.1.0",
4   "env": "development",
5   "uptimeSeconds": 226,
6   "requestId": "a5b38b22-b8f2-482f-b00c-25d8091cbf20"

```

```

> resumes-ai-rag@0.1.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts

[INFO] 22:07:14 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
Server listening on port 3000
{"level":30,"time":177990053676,"pid":22776,"hostname":"3e485cdec636a0e","route":"/v1/health%20%20","method":
GET,"requestId":"f6df1f96-068f-4a5f-a9f7-d2234c41f09c","msg":"request.start"}
{"level":30,"time":177990053779,"pid":22776,"hostname":"3e485cdec636a0e","route":"/v1/health%20%20","method":

```

PROCEED WITH MONGODB HEALTHCHECK

HTTP <http://localhost:3000/v1/health/db> Save Share

GET <http://localhost:3000/v1/health/db> Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings [Cookies](#)

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results 200 OK 8 ms 352 B ⌵ ⋮

{ } JSON Preview Visualize ⌵

```

1 {
2   "ok": true,
3   "pingMs": 2,
4   "measuredMs": 2,
5   "requestId": "bfec3fc9-0ce6-4be5-b3ce-b3bbe1dd29ca"
6 }

```

> Finalize and update TODOs (7/7) ⌵

+ ⚙️ .env.example x ⚙️ .env x ⚡ Architecture.md

"To implement Vector Embedding +
`/v1/embeddings` with Architecture.md and
create_endpoint.md prompt file. Confirm if you
have everything to code (Do not generate code)"

+ ? | GPT-5 mini · Medium | ↕️ ⬆️

COUPLE OF QUERIES ANSWEREED IN ASK MODE

Q: Do you want to use `node-fetch`/built-in `fetch` or
`axios` for the Mistral HTTP calls?
A: 1

Q: Confirm embedding behavior
A: CHOOSE WHICHEVER IS BETTER

Q: Preferred retry and timeout policy for Mistral calls
A: CHOOSE WHICHEVER IS BETTER

ADD MISTRAL MODEL AND EMBEDDING TO .ENV FILE

```

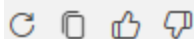
EMBEDDING_MODEL=mistral-embed
EMBEDDING_DIM=1024

```

SWITCH TO AGENT MODE TO GENERATE CODE TO IMPLEMENT VECTOR EMBEDDINGS

- Uses built-in `fetch`. If your Node.js version lacks global fetch, run Node 18+ or install a polyfill (node-fetch) and update the service.
- Mistral endpoint is configurable via `MISTRAL_EMBED_URL` (defaults to <https://api.mistral.ai/v1/embeddings>). Key and model read from `.env`.
- Tests mock the service; you can run them with `npm test`.

Would you like me to run the test suite now?



GPT-5 mini • 1x

> Finalize and update TODOs (8/8)



> 5 files changed +149 -4

Keep

Undo



```
> resumes-ai-rag@0.1.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
```

```
[INFO] 22:25:47 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
Server listening on port 3000
```

POST

http://localhost:3000/v1/embeddings

Send

Docs

Params

Authorization

Headers 9

Body

Scripts

Tests

Settings

Cool

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautif

```
1 {"input":"hello world"}
```

Body

Cookies

Headers 7

Test Results

200 OK • 9.92 s • 19.21 KB •

{ } JSON

Preview

Visualize

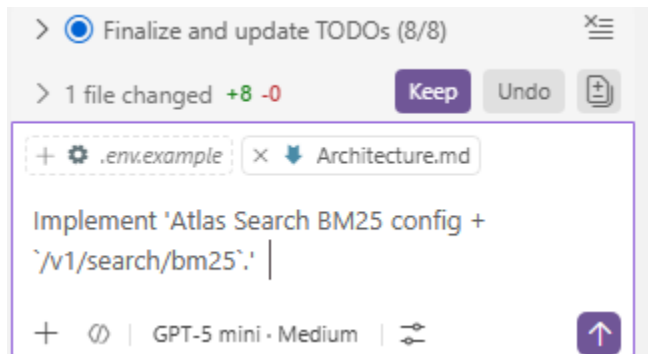
```

1 {
2   "requestId": "21340f4e-c71e-407d-b60b-3e447f4bb1cd",
3   "model": "mistral-embed",
4   "embedding": [
5     -0.01922607421875,
6     0.0268096923828125,
7     0.01203155517578125,
```

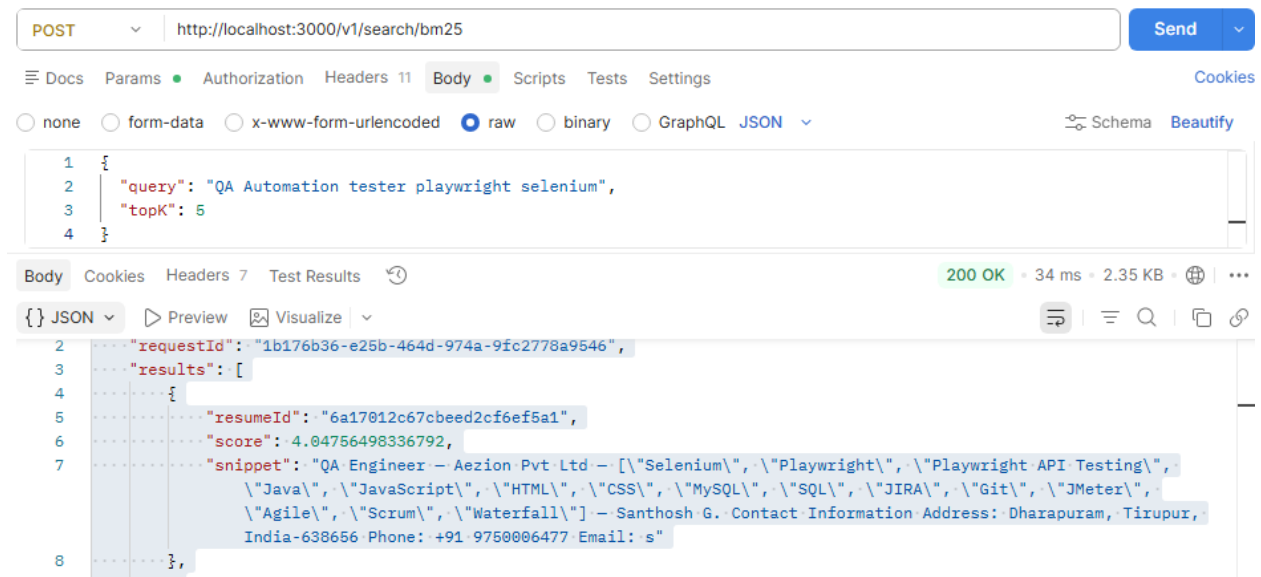
LETS UPDATE THE GROQ MODEL AND KEY IN THE ENV FILE AND SWITCH TO AGENT MODE

```
15 GROQ_API_KEY=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
16 GROQ_LLM_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
```

LETS IMPLEMENT ATLAS SEARCH ENDPOINT



LETS TEST THE ENDPOINT THROUGH POSTMAN



LETS IMPLEMENT HYBRID SEARCH

http://localhost:3000/v1/search/hybrid Save Share

POST http://localhost:3000/v1/search/hybrid Send

Docs Params Authorization Headers 11 Body Scripts Tests Settings Cookie

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```

1 {
2   "query": "QA Automation tester playwright selenium",
3   "topK": 5
4 }

```

Body Cookies Headers 7 Test Results 200 OK 2.35 s 4.41 KB ⌕ ⋮

{ } JSON Preview Visualize ⌕ ⌕ ⌕ ⌕ ⌕

```

8 {
9   "resumeId": "6a17012c67cbeed2cf6ef577",
10  "score": 3.2817251682281494,
11  "snippet": "Test Lead - HCL Technologies - [\"Selenium WebDriver\", \"Playwright\", \"TypeScript\",
12            \"JavaScript\", \"GitHub\", \"Core Java\", \"REST API\", \"Postman\", \"HP Quality Center\", \"JIRA\",
13            \"Agile\", \"Data Migration Testing\"] - 1 Confidential C Muthu P +91-9123970724 muthusalem@gmail.com
14            Experience Summary: Test Lead "
15 }

```

```

"vector": [
  {
    "resumeId": "6a17012c67cbeed2cf6ef57d",
    "score": 0.8383101673184077,
    "snippet": "Test Analyst - Tata Consultancy Services -
      \"JavaScript\", \"TestNG\", \"Postman\", \"Agile\"
      \"Jenkins\", \"Maven\", \"Perfecto\", \"Adobe Expe:
      Email: manoj.mb@gmail.com Mobile: (+91) 9597123248
  },

```

LETS IMPLEMENT RERANK

- If you want pretty JSON output, pipe through

jq:

curl ... | jq

Would you like me to run the test suite now to confirm everything is clean, or do you want me to run a live curl against your dev server?



GPT-5 mini • 1x

give me sample endpoint for rerank

HTTP <http://localhost:3000/v1/search/rerank> Save Share

POST <http://localhost:3000/v1/search/rerank> Send

Docs Params Authorization Headers 11 **Body** Scripts Tests Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```

1 {
2   "query": "QA Automation tester playwright selenium",
3   "candidates": [
4     {
5       "resumeId": "6a17012c67cbeed2cf6ef5a1",
6       "snippet": "QA Engineer – Playwright, Selenium, API testing, Java",

```

Body Cookies Headers 7 Test Results 200 OK • 203 ms • 571 B •

{{ JSON Preview Visualize

```

4 {
5   "resumeId": "6a17012c67cbeed2cf6ef5a1",
6   "snippet": "QA Engineer – Playwright, Selenium, API testing, Java",
7   "score": 3.2
8 },
9 {
10  "resumeId": "6a17012c67cbeed2cf6ef577",
11  "snippet": "Test Lead – Selenium WebDriver, Playwright, TypeScript",

```

LETS IMPLEMENT SUMMARIZE ENDPOINT

+ TS LLMService.ts ✕ Architecture.md

Implement 'Summarization config +
`/v1/search/summarize`.'

+ GPT-5 mini · Medium ↑

HTTP <http://localhost:3000/v1/search/summarize> Save Share

POST <http://localhost:3000/v1/search/summarize> Send

Docs Params Authorization Headers 11 **Body** Scripts Tests Settings Co

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beaut

```

2 "query": "QA Automation tester with Playwright and Selenium experience",
3 "candidate": {
4   "resumeId": "6a17012c67cbeed2cf6ef5a1",
5   "snippet": "QA Engineer – Playwright, Selenium, API testing, JavaScript, Java – 7+ years experience leading
6   automation efforts..."

```

Body Cookies Headers 7 Test Results 200 OK • 185 ms • 503 B •

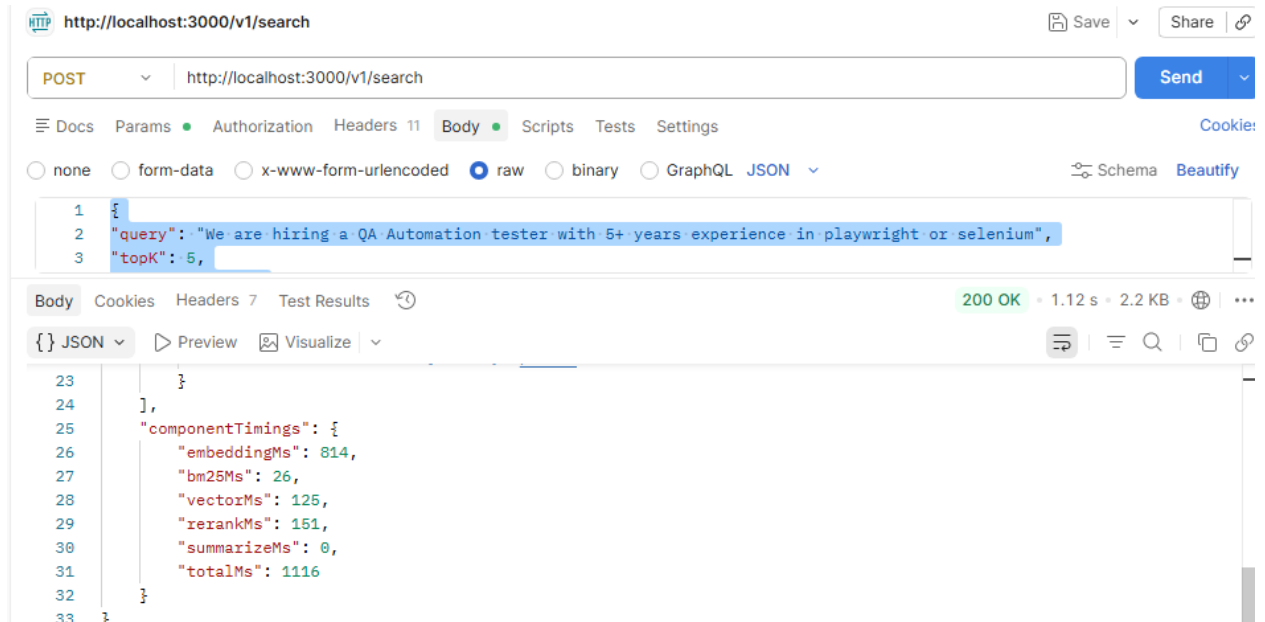
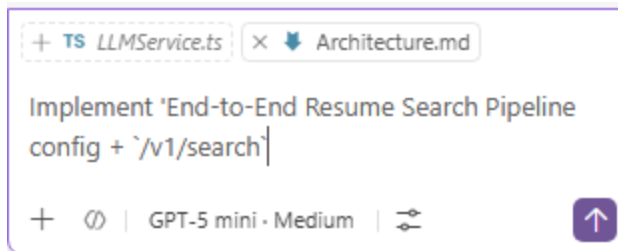
{{ JSON Preview Visualize

```

1 {
2   "requestId": "27857c03-e22a-4e6a-9816-5c1b0d6611f8",
3   "summary": "QA Engineer – Playwright, Selenium, API testing, JavaScript, Java – 7+ years experience leading
4   automation efforts...",
5   "model": "meta-llama/llama-4-scout-17b-16e-instruct"

```

LETS IMPLEMENT THE FINAL E2E RESUME SEARCH



```
{
  "query": "We are hiring a QA Automation tester with 5+ years experience in playwright or selenium",
  "topK": 5,
  "rerankTopK": 5,
  "summarize": false
}
```

Response

```
{
  "requestId": "f1ec608e-6093-4e31-91e2-b5710a076ff6",
  "results": [
    {
```

"resumeld": "6a17012c67cbeed2cf6ef577",

"snippet": "Test Lead — HCL Technologies — [\"Selenium WebDriver\", \"Playwright\", \"TypeScript\", \"JavaScript\", \"GitHub\", \"Core Java\", \"REST API\", \"Postman\", \"HP Quality Center\", \"JIRA\", \"Agile\", \"Data Migration Testing\"] — 1 Confidential C Muthu P +91-9123970724 muthusalem@gmail.com Experience Summary: Test Lead "

},

{

"resumeld": "6a16c94d67cbeed2cf6ef55f",

"snippet": "Qeagle Assurance Private Limited Address: Tower C 2nd Floor, TEK Meadows Campus, No.51, Rajiv Gandhi Salai, Sholinganallur, Chennai – 600119. CIN - U72900TN2019PTC132090 info@qeagle.com | www.qeagle.c"

},

{

"resumeld": "6a17012c67cbeed2cf6ef57d",

"snippet": "Test Analyst — Tata Consultancy Services — [\"Selenium\", \"Playwright\", \"Java\", \"JavaScript\", \"TestNG\", \"Postman\", \"Agile\", \"Scrum\", \"Waterfall\", \"Git\", \"GitHub\", \"Jenkins\", \"Maven\", \"Perfecto\", \"Adobe Experience Manager\", \"Pega Framework\"] — Manoj G Email: manoj.mb@gmail.com Mobile: (+91) 9597123248 "

},

{

"resumeld": "6a17012c67cbeed2cf6ef5a1",

"snippet": "QA Engineer — Aezion Pvt Ltd — [\"Selenium\", \"Playwright\", \"Playwright API Testing\", \"Java\", \"JavaScript\", \"HTML\", \"CSS\", \"MySQL\", \"SQL\", \"JIRA\", \"Git\", \"JMeter\", \"Agile\", \"Scrum\", \"Waterfall\"] — Santhosh G. Contact Information Address: Dharapuram, Tirupur, India-638656 Phone: +91 9750006477 Email: s"

},

{

"resumeld": "6a17012c67cbeed2cf6ef5bd",

"snippet": "Automation Tester — LTIMindtree Ltd. — [\"Selenium WebDriver\", \"Leapwork\", \"ServiceNow ATF\", \"AWS\", \"Cucumber\", \"TestNG\", \"Maven\",

\"Jenkins\", \"GitHub\", \"JIRA\", \"HP ALM\", \"Qtest\", \"Core Java\"] — GOPI L Test
Automation Gopi324@gmail.com (+91) – 8939680229 OBJECTIVE To obtain a Software QA
Analyst/Teste"

}

],

"componentTimings": {

"embeddingMs": 814,

"bm25Ms": 26,

"vectorMs": 125,

"rerankMs": 151,

"summarizeMs": 0,

"totalMs": 1116

}

}